Matthew Frieri

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Education

University of Toronto | *CGPA: 3.95/4* Graduating Spring 2028

- Computer Science Major and Data Science Specialist
- Focus in Technology Leadership. Chosen as one of 30 from 500+ applicants.

Experience

Software Developer | *RBC Capital Markets* Jan – August 2026

- Developed a pub-sub framework to serve real-time inference of the volume prediction model mentioned below. The signal is used by RBC's algorithmic trading platform, Aiden, which trades ∼**$1.54B of daily notional value**.
- Configured **Prometheus** and **Loki** data sources to create a **Grafana** dashboard for the volume signal, monitoring live performance and alerting unwanted behavior via email. Critical to the successful deployment to **production**.
- Developed a framework to abstractly **read/write/process data in parallel** via multi-threading and multi-processing from various sources to sinks (parquet, arrow, s3, kdb+). Queried data up to **53% faster in parallel**.

ML Engineer | *RBC Capital Markets* May – Dec 2025

- Developed a **neural network** to predict volume in the global equities market. Consolidated multiple legacy models, expanding inference coverage by **43% more stocks worldwide**, while **reducing aggregate error by 21%**.
- Queried time series data in kdb+/Q and conducted **feature engineering** in Python by analyzing feature/target distributions, applying appropriate **normalizations**, and creating meaningful **categorical embeddings**.
- Built a Streamlit dashboard to monitor feature importance in real-time, supporting model refinement through the ablation of 159 features, **reducing model size by 64%** while maintaining symmetric and asymmetric error.

Full Stack Developer | *MultiCloud Decentralized Applications Inc.* July – Dec 2024

- Developed and designed the front end of the company's website using **Next.js** and **Tailwind CSS**, employing modular design and reusable components to streamline development and significantly boost efficiency.
- Implemented backend features including a tree based file system, file search, and an automated email service, helping bring the platform closer to production readiness.
- Enhanced **SEO** and internal website performance using Google Lighthouse, realizing a **40% increase** in web crawler visibility, loading times, and accessibility.

Projects

FPGA Placement and Routing in Minecraft | *Python* Oct 2025 – Present

- Created a Verilog-like language that gets interpreted, synthesized, and placed onto a **custom island-style FPGA** built in Minecraft. The board features configurable logic blocks, routing channels, switch boxes, and I/O blocks.
- Developed a **multi-objective cost function** balancing propagation delay and wire length for placement/routing.
- Applied heuristic algorithms to place and route netlists onto the FPGA, successfully placing **20% more circuits** than naive methods. Currently exploring ML approaches to further improve success rate and routing metrics.

Cash App – Exposing Bias in Fraud Detection | *Python, Statsmodels, Flask, React* Sept – Dec 2024

- Worked with a team of 5 in collaboration with Cash App to develop a dashboard that reveals bias in their AI fraud detection system, identifying disproportionate targeting through detailed, interpretable visualizations.
- Trained an **ARIMA** model to predict the impact of removing specific biases from Cash App's fraud detection system, estimating a **20% reduction in false positives** for underrepresented user groups.

Impactify (1st Place @ Hack The Valley) | *Esp32, Django, PostgreSQL, React, Twilio* Oct 2024

- Embedded hardware in a hockey helmet to detect high impact collisions so coaches can monitor their players live.
- Developed a live dashboard for coaches and players, where data is streamed from the helmet.
- Implemented automatically sending SMS notifications to coaches when a player experiences a traumatic impact.

Technical Skills & Interests

Programming Languages: Python, C++, C, kdb+/Q, Java, SQL, Verilog, JavaScript, TypeScript, HTML/CSS
Technologies: Git, PyTorch, MLFlow, Pandas, Polars, Numpy, S3, React, Django, FastAPI, OpenCV, MongoDB
Interests: Outdoor Bouldering, Backcountry Canoe Camping, Tennis, Table Tennis, Self Hosting / Home Server